

Week 2 – Workshop Plan

Physical Properties, Mixing Laws, and Asking the Right Questions

Geophysics 3A: Potential Fields & Geothermics

February 26, 2026

Duration: 3 hours

Setting: Lecture room (no lab instruments)

Purpose

This workshop develops intuition for how physical properties (density ρ , thermal conductivity k , thermal diffusivity κ , bulk modulus K_T , shear modulus μ , seismic velocities V_P , V_S , magnetic susceptibility χ , and heat production A) depend on state variables (pressure P , temperature T , composition, porosity/cracks), and how geometry and fabric determine appropriate mixing relations (arithmetic, geometric, harmonic; Voigt–Reuss–Hill). Students practice cross-disciplinary collaboration and formulate the key questions a practicing geophysicist must ask a geologist before modeling or surveying.

Learning Outcomes

- Select and justify mixing laws based on rock fabric (random granular, layered/foliated, fractured) and target property (bulk vs. transport vs. elastic).
- Reason about how P – T –composition/porosity shift properties and create anisotropy.
- Identify which properties most strongly influence gravity, magnetic, and conductive heat-flow interpretations.
- Elicit geological information via well-posed questions to constrain non-uniqueness.

Reference concepts are aligned with the Week 2 notes on physical properties and mixing relationships.

Roles and Grouping

Create mixed groups (4–5 students each) with both geology and physics/engineering backgrounds. Rotate roles: *Facilitator*, *Unit/Assumption Checker*, *Recorder*, *Skeptic*.

Session Flow (180 min)

0–10 min: Warm-up “What do we need to know?”

Individually list three properties needed before interpreting gravity, magnetics, or heat flow, and for each list two controls (e.g., P , T , composition, porosity) and one estimation method (measure, proxy, or model). Share in pairs, then with group.

10–25 min: Micro-lecture (mini deck)

- Bulk-additive vs. pathway-limited vs. elastic-interaction properties.

- Mixing relations: arithmetic (parallel), harmonic (series), geometric (random granular), Voigt–Reuss–Hill (elastic bounds).
- Scale-dependent anisotropy (micro/meso/macro fabrics).

25–40 min: Live Demonstration — “Crowd Conductivity”

Aim: Intuitively illustrate why different mixing laws arise and how anisotropy appears.

1. **Random scattering** (students distributed irregularly): volunteer transits across room. Multiple semi-obstructed pathways $\Rightarrow$ geometric-like effective behavior.
2. **Parallel corridors** (lanes aligned with motion): fastest transit $\Rightarrow$ arithmetic mean (parallel pathways).
3. **Series barriers** (rows perpendicular to motion): slowest transit $\Rightarrow$ harmonic mean (series resistances).
4. **Anisotropy**: compare transit parallel vs. perpendicular to fabric.

Optional: time a handful of transits and note distribution shape differences.

45–60 min: Photo-Based Anisotropy

(micro–meso–macro classification)

Distribute a set of photos (thin section, hand sample/outcrop close-up, outcrop/aerial). For each image, groups identify: geometry (random granular/layered/foliated/fractured), appropriate mixing law for density, conductivity, and moduli, whether the material is isotropic at that scale, and how conclusions change with scale.

60–120 min: Scenario Rotations (rich, real-world cases)

Each group tackles two scenarios (20 min each: 12 min discussion, 5 min notes, 3 min question drafting). Use the dedicated scenario handouts. Outcomes must include: (i) property/mixing-law decisions and (ii) 2–3 *questions to ask the geologist* before committing to a model/survey.

120–135 min: Scenario Debrief

135–165 min: Optional Simulation Walkthrough

Show a pre-built Python visualization of transit times through three media (random, parallel, series) and the resulting histogram shapes. Relate to effective properties and mixing choices.

165–180 min: Q & A

Discussion of any new or unresolved questions.

Materials

- Scenario worksheet
- Photo worksheet
- Slide deck for micro-lecture and activity instructions.
- Optional: Python simulation script (with figures saved to PNG) for in-class demo or homework exploration.

Assessment

- Scenario notes (individual or group, 1–2 pages) and question list contribution.

- Photo activity classification sheet.
- Exit ticket reflection.